

# Mert Koca

Mapping neural circuits at nanometer scale, from bench to supercomputer.

LOCATION  
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Highly self-driven and intellectually curious individual with a strong foundation in machine learning and neuroscience, working at the intersection of expansion-microscopy connectomics and deep-learning segmentation. Skilled at synthesizing complex concepts across disciplines into practical, innovative solutions while managing long-term projects with a balance of creativity and structured execution.

## EXPERIENCE

Jul 2025 - Present

Istanbul, Turkey

### Research Intern

KUTTAM, Koç University Research Center for Translational Medicine

#### FLAGSHIP RESEARCH

- Studied various **Expansion Microscopy (ExM)** protocols in the lab (ProExM, Magnify, and single-shot 20× ExM) to surpass the optical diffraction limit, enabling nanometer-scale imaging of neural tissue on confocal microscopes.
- Adapted and ran deep-learning **volume-segmentation pipelines** (Flood-Filling Networks, U-Net variants) for neural and vascular structures, working toward computer-based 3D brain mapping.
- Learned and operated **high-performance computing** on TRUBA (Turkey's national HPC center): SLURM batch scheduling, multi-GPU (JAX) training, and Apptainer/Singularity containers.

Oct 2024 - Feb 2026

Scholarship to Jul 2025

### Embedded Machine Learning Scholar

TUSEB · Health Institutes of Turkey

- Engineered a real-time **embedded machine learning** pipeline for EMG-based gesture recognition, from DSP filter design to on-device feature extraction and inference.
- Trained models in Python and deployed them as **Int8-quantized TensorFlow Lite Micro** models running on an ESP32-S3 microcontroller for real-time inference.
- Produced data-analysis visualizations and animations in Python (Manim).

2024 - 2025

Quarter-finalist

### Member

Easyegitim · İTÜ Çekirdek

- Combined **agentic tools** with specialized capabilities for students to achieve maximum efficiency from personalized education.

Jan 2024 - Jul 2024

National finalist

### Team Member

Tez-Türk

- EMG gesture classification** on the Ninapro DB5 dataset; built an MLP classifier for a prosthetic hand.

## SELECTED PROJECTS

**mertoshi.online** *Personal website & agentic RAG platform, solo, end-to-end*

- Frontend:** React 19 + TypeScript + Vite + Tailwind + Spline 3D, deployed on Azure Static Web Apps.
- Backend:** Python Azure Functions RAG using Azure OpenAI "On Your Data" + Azure AI Search (hybrid vector/semantic), secured with Managed Identity + Azure Key Vault.
- Features a custom AI assistant ("Lundo") and a document-intelligence + semantic-chunking indexing pipeline.
- Turned my bachelor thesis into animated, illustrated blog chapters, and translated what I learn in the lab into ongoing blog posts.

ESP32-S3 Bionic Hand *Real-time EMG-controlled bionic hand*

- Recognizes **11 hand gestures** in real time from **6 EMG sensors** (MyoWare) to drive a bionic prosthetic hand.
- On-device ML with **TensorFlow Lite Micro** on the ESP32-S3: Int8-quantized MLP, TD4 EMG features, and real-time IIR filtering in C/C++, under 300 ms end-to-end latency.

liconn-ffn-truba *Open-source HPC connectomics pipeline*

- Trains Google's **Flood-Filling Network** (JAX) on real expansion-microscopy connectomics data, scaling 1 → 4 GPUs on TRUBA.
- Public GitHub repository with an animated report.

EDUCATION

B.Sc., Molecular Biotechnology

Turkish-German University · Beykoz, Istanbul · *Graduated*  
Jun 2026 · GPA 3.14 / 4.0

Bachelor thesis (in German): light-microscopic connectomics beyond the diffraction limit: expansion microscopy and AI-assisted segmentation of neural networks. Advisor: Prof. Dr. Atay Vural.

CERTIFICATION

Foundations of Quantum Mechanics

University of Colorado Boulder · *Coursera* · [Verify](#)

LANGUAGES

**Turkish:** Native · **German:** Advanced (C1, medium of instruction) · **English:** Fluent

COURSES

Quantum Biology

Onur Pusuluk · *Bilimler Köyü (Science Village)* · 1 week

Advanced Methods in Machine Learning

Çağatay Yıldız, İlker Birbil, Özgür Martin, Tolga Birdal & Sinan Yıldırım · *Bilimler Köyü (Science Village)* · 1 week

PUBLICATIONS

- Coskuner-Weber O., Koca M., Uversky V. N. “Molecular Crowding by Computational Approaches.” Springer Nature. DOI: 10.1007/978-3-032-03370-3\_21
- Koca M. “Kuantum-Klasik Bilinç Modelleri ve İnterdisipliner Perspektifler.” [Quantum-Classical Models of Consciousness and Interdisciplinary Perspectives] Quantum Dergisi, Issue 5 (qturkey.org).
- Third manuscript in preparation.

SKILLS

|                  |                                                                                                                                 |
|------------------|---------------------------------------------------------------------------------------------------------------------------------|
| PROGRAMMING      | Python, TensorFlow, JAX, NumPy, PyTorch (learning), Bash                                                                        |
| MACHINE LEARNING | Deep learning (CNN, LSTM, MLP, Flood-Filling Networks, U-Net), Random Forest, Gaussian Processes, LangChain / LangGraph, RAG    |
| HPC & CLOUD      | TRUBA HPC, SLURM, multi-GPU training, Apptainer / Singularity, Azure (Functions, AI Search, OpenAI), Git / GitHub               |
| IMAGING & BIO    | Expansion microscopy, connectomics, image segmentation, pan-protein staining, immunofluorescence staining, vibratome sectioning |
| TOOLS & VIZ      | Manim, Matplotlib                                                                                                               |

INTERESTS

Photography (street & portrait) · running · interdisciplinary science · behavioral neuroscience.